Year 8 Science – Booklet 13: Physics

Forces in Action (Friction and Terminal Velocity) — ANSWERS

Note: this booklet's content is identical to the “Friction and Terminal Velocity” section of Year 8 Booklet 4, so the same worked answers apply.

Quick Test (p.93)

1. In the opposite direction to the object's motion (or attempted motion) – it opposes movement.
2. Shaping an object so it cuts through air (or water) with less resistance, reducing drag.
3. A substance (e.g. oil or wax) used to reduce friction between two surfaces in contact.
4. Any valid example, e.g. friction between tyres and the road lets a car grip, steer and brake safely; friction between shoes and the ground stops you slipping when walking.
5. It travels at a constant speed – its terminal velocity, if falling/moving through a fluid.
6. Heating (a rise in temperature) and wearing away (wear) of the surfaces in contact.

Section A – Multiple Choice (5 marks)

1. b) temperature
2. b) roughening the surfaces
3. a) having a streamlined shape
4. a) increase
5. d) terminal velocity

Section B (12 marks)

1. Fill in the missing words (10 marks)

At the start of a bobsleigh race the team has to push the bobsleigh as hard as they can. The team members wear **spikes** on the bottoms of their shoes to prevent them from slipping. As the sleigh gains **speed** the team jump on board and crouch down so that they are **streamlined**. As the bobsleigh travels down the run it gains speed. The **frictional** forces between the sleigh and the **ice** and between the runners and the **ice** increase. To keep these forces to a minimum the runners are coated with a **lubricant** such as **wax** and the shape of the sleigh is **streamlined**.

2. Advantage/disadvantage of friction in sport

a). Advantage: e.g. rock climbing – friction between hands/shoes and the rock lets you grip and avoid slipping.

b). Disadvantage: e.g. speed skating/cycling – too much friction (surface or air resistance) slows the athlete down.

Section C (9 marks)

1. Skydiver falling

a). Gravity (her weight).

b). It increases – she accelerates.

c). Air resistance (drag).

d). They are balanced (air resistance equals her weight), so there's no resultant force and she falls at a constant speed.

e). Opening the parachute greatly increases her surface area, which greatly increases air resistance. This becomes much bigger than her weight, creating a net upward force that decelerates her until a new, lower terminal velocity is reached.

2. Bicycle brake blocks

a). They get hot (from friction), and they wear away/wear down over time.

b). The chain (also accept: the wheel axles/bearings, gears).